

Lesson 4 Sets

Set : a collection of objects

- unordered $\{1, 2\} = \{2, 1\}$
- nonrepetitious $\{1, 1\} = \{1\}$

$$x \in A$$

x is an element of set A

$$1 \in \{1, 2\}, \quad 3 \notin \{1, 2\}$$

not an element

Number sets

$\mathbb{R}$: set of reals (real #'s)

$\mathbb{Z}$: set of integers

Def An integer x is even if $x = 2n$ for some $n \in \mathbb{Z}$.

2 ways to display/define elements of a set :

① List the elements ($\{0, 1, 2, 3, \dots\}$)

$\mathbb{N}$ = set of natural #'s
= positive integers
= $\{1, 2, 3, \dots\}$

② Use set-builder notation

$$\mathbb{N} = \{ \underbrace{x \in \mathbb{Z}}_{\text{variable}} : \underbrace{x > 0}_{\text{attribute}} \}$$

~~$\pi/2$~~

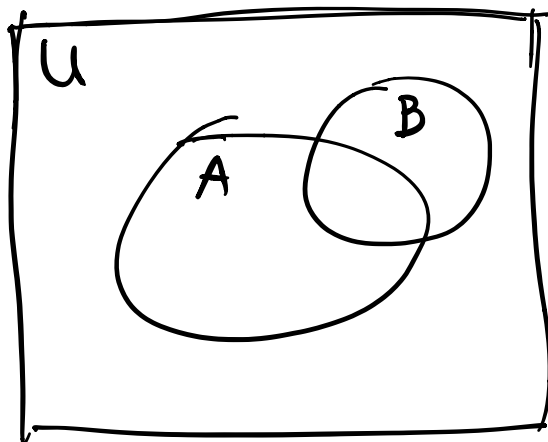
Rational numbers $\mathbb{Q}$

$$\mathbb{Q} = \{ \frac{m}{n} : m, n \in \mathbb{Z}, n \neq 0 \}$$

↑

form | attributes
"such that"

Def Set A is a subset of set B, $A \subseteq B$,
if every element of A is an element of B.



Venn diagram

U : universal set

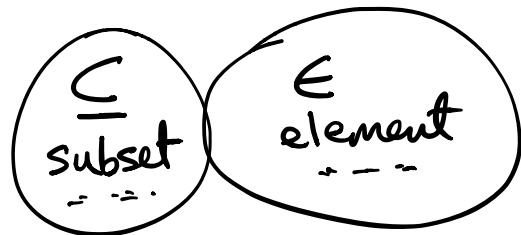
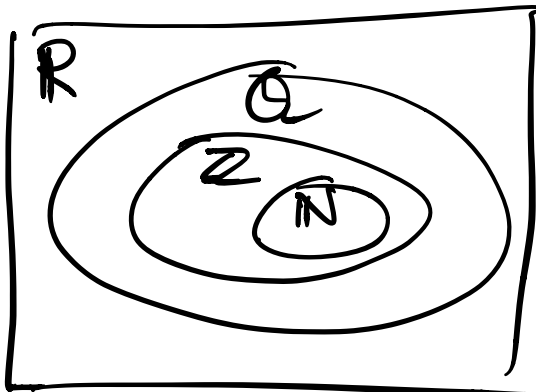
$$A \subseteq U$$

↑
any set

$\emptyset$: empty set

$$\emptyset \subseteq A$$

↑
any set



Power set of A

$\mathcal{P}(A)$ = set of all subsets of A

Ex $A = \{1, 2\}$

$$\mathcal{P}(A) = \{ \emptyset, \{1\}, \{2\}, A \}$$

$$1 \in A \checkmark$$

$$\{1\} \in A \times$$

$$\{1\} \subseteq A \checkmark$$

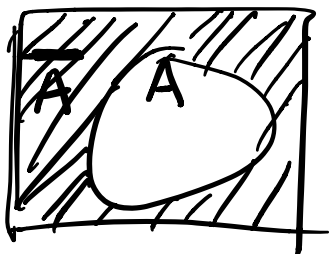
$$\{\{1\}\} \in A \times$$

$$\{1\} \in \mathcal{P}(A) \checkmark$$

$$\{1\} \subseteq \mathcal{P}(A) \times$$

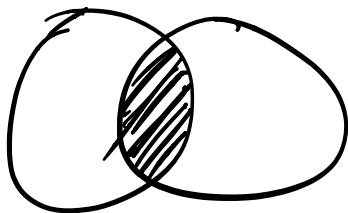
$$\{\{1\}\} \subseteq \mathcal{P}(A) \checkmark$$

- Set operations A, B : sets (number operations)
 $+$, $\times$
 (logical operations)
 connectives
 "and", "or"
- complement of A
 $\bar{A} = \{x \in U : x \notin A\}$



"negation"

- intersection of A and B



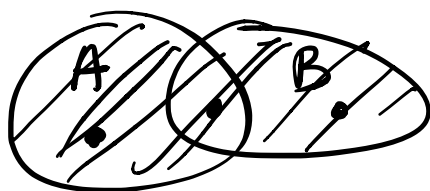
$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

$x \in U$ "and"

- Union of A and B

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

inclusive



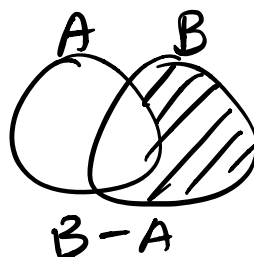
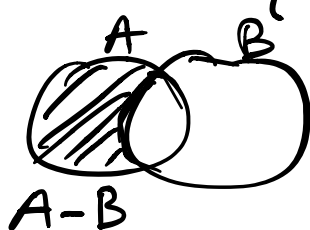
"or"

A, B disjoint : $A \cap B = \emptyset$



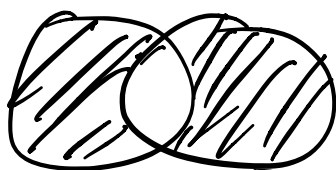
- difference of A and B

$$A - B = \{x : x \in A \text{ and } x \notin B\}$$



- Symmetric difference of A and B

$$A \Delta B = (A - B) \cup (B - A)$$



"xor"

$$U = \{1, 2, 3, \dots, 9\} \quad A = \{2, 4, 6, 8\}, \quad B = \{3, 6, 9\}$$

$$\bar{A} = \{1, 3, 5, 7, 9\}, \quad A \cap B = \{6\}, \quad A \cup B = \{2, 3, 4, 6, 8, 9\}$$

$$A - B = \{2, 4, 8\}, \quad B - A = \{3, 9\}$$

$$A \Delta B = \{2, 3, 4, 8, 9\}$$

- Cartesian product of A and B

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$$

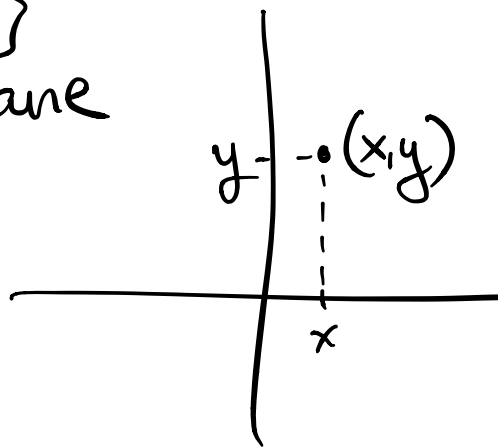
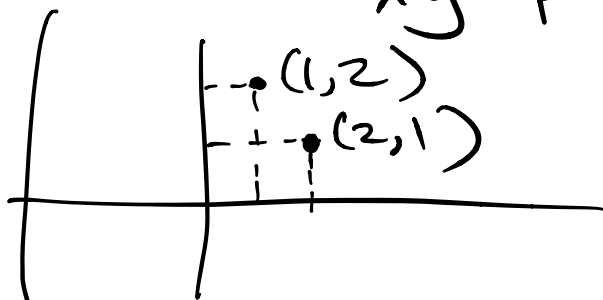
$$\textcircled{2} \quad A = \{1, 3\}, \quad B = \{2, 4, 6\} \quad \textcircled{3}$$

pair

$$A \times B = \{(\underline{1, 2}), (1, 4), (1, 6), (3, 2), (3, 4), (3, 6)\}$$

$$B \times A = \{(\underline{2, 1}), (2, 3), (4, 1), (4, 3), (6, 1), (6, 3)\}$$

$\mathbb{R}^2$: Cartesian plane
xy-plane



$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$$

$$= \{(x, y) : x, y \in \mathbb{R}\}$$

$$A = \{x \in \mathbb{R} : \underline{-2 \leq x \leq 3}\} = [-2, 3]$$

$$A \times \mathbb{R} = \{(x, y) : x \in A \text{ and } y \in \mathbb{R}\}$$

interval

